

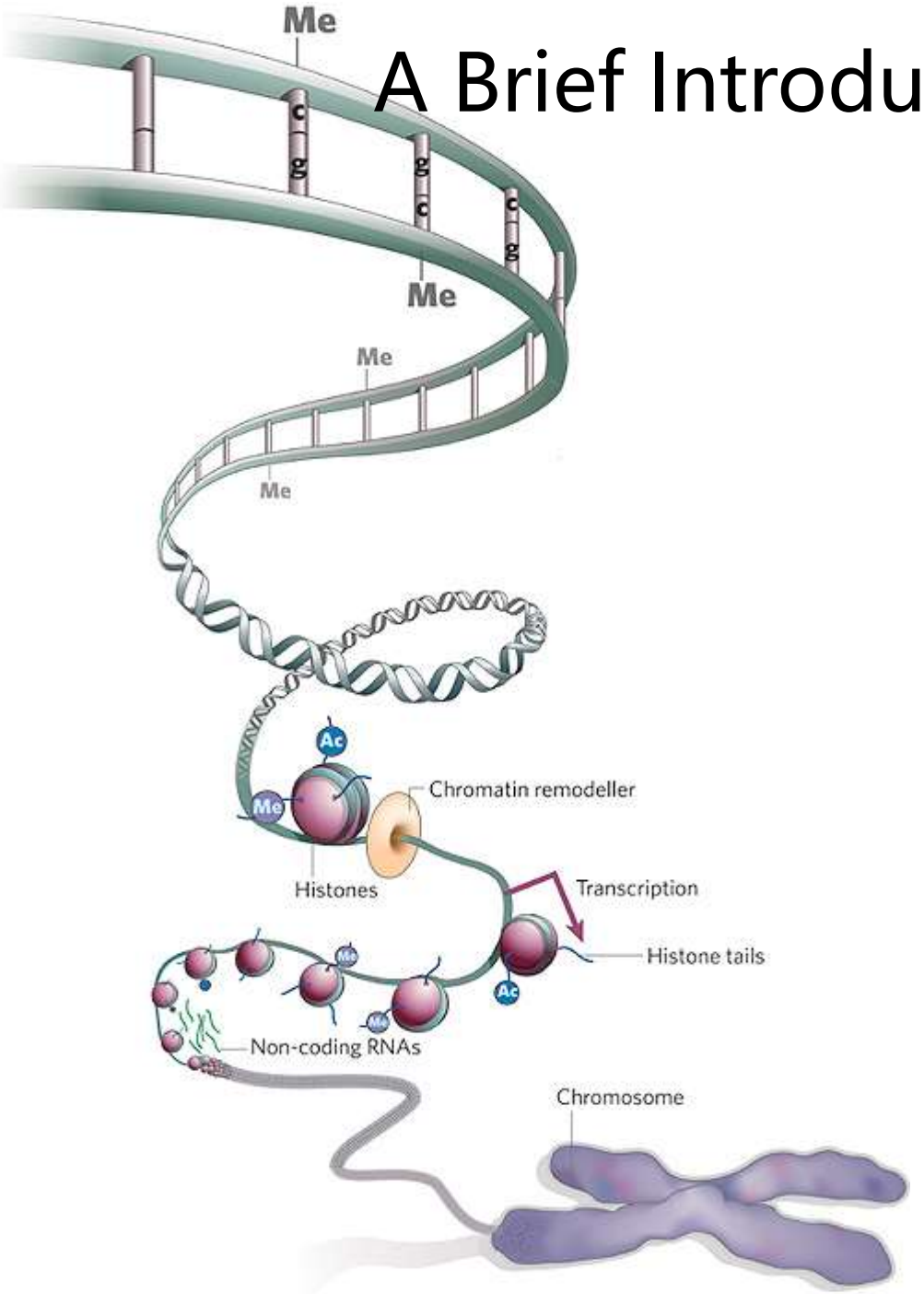
高通量测序技术及其应用

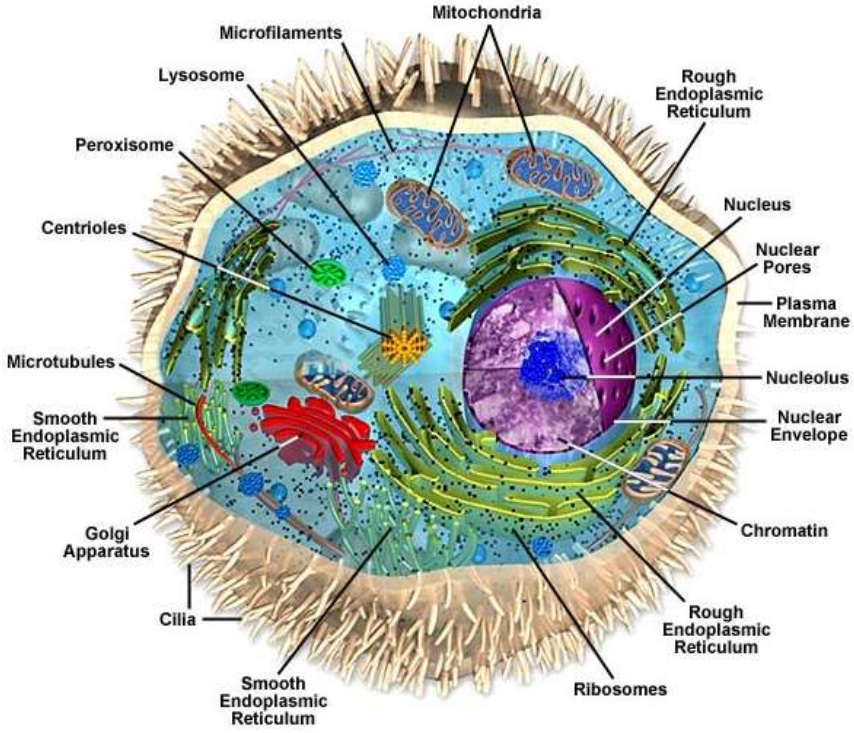
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中国科学院北京生命科学研究院
计算生物学联合研究中心

2017.11

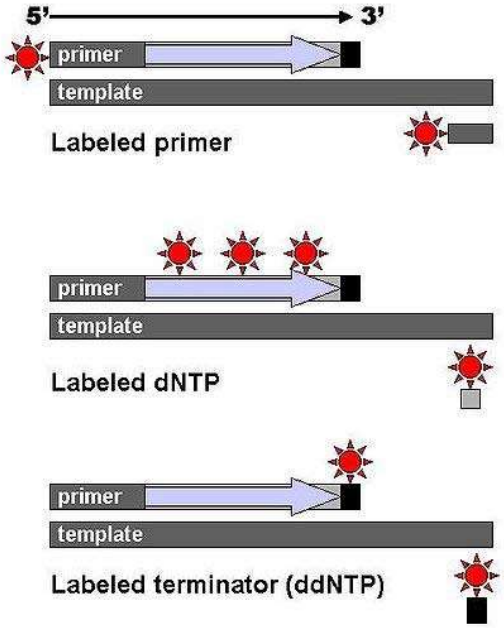
A Brief Introduction to Genomics

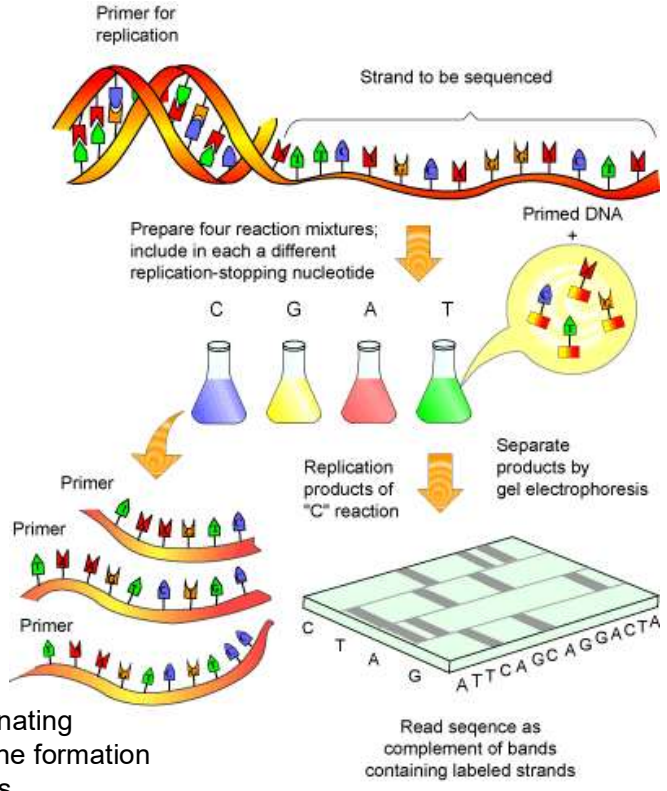






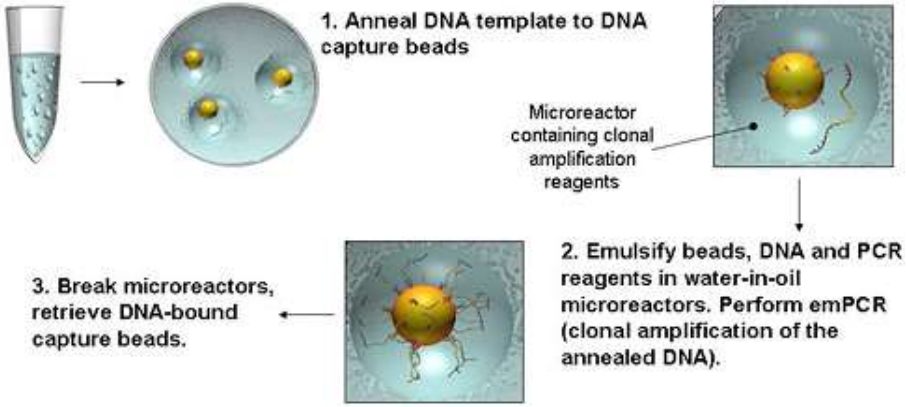
Sanger sequencing

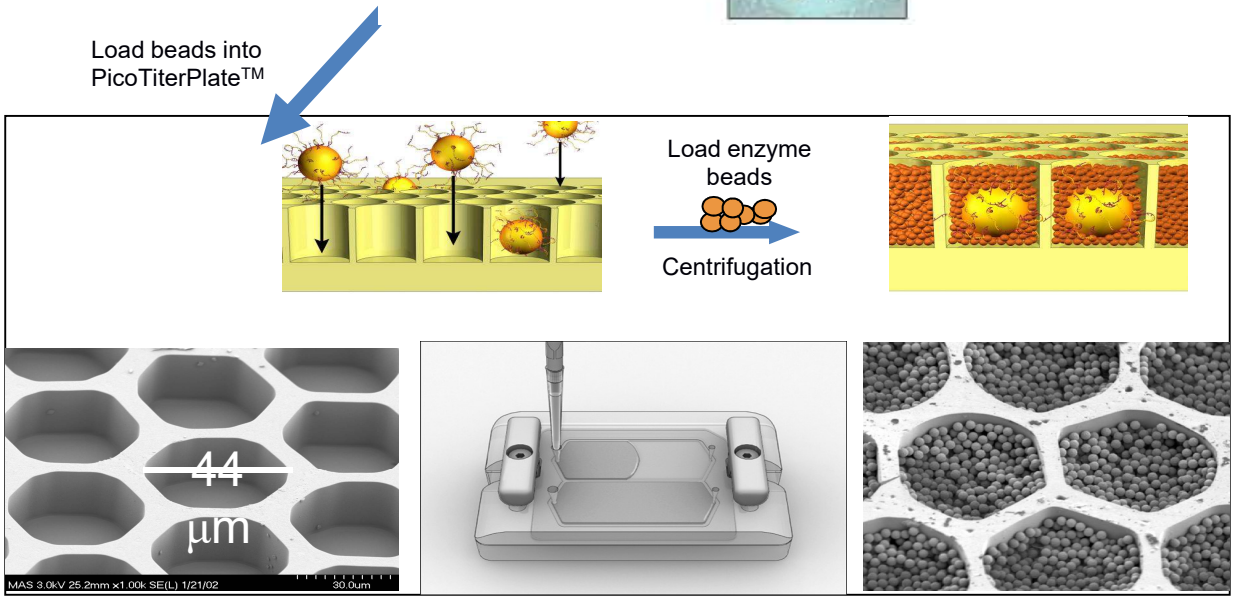




To each reaction is added **only one** of the four dideoxynucleotides which are the chain-terminating nucleotides, **lacking a 3'-OH group** required for the formation of a phosphodiester bond between two nucleotides, thus **terminating DNA strand extension** and resulting in various DNA fragments of varying length.

High throughput sequencing





High Throughput Sequencing Technologies



AB 3730xl



Roche 454



Illumina Hiseq



SOLiD



MiSeq



Ion Torrent



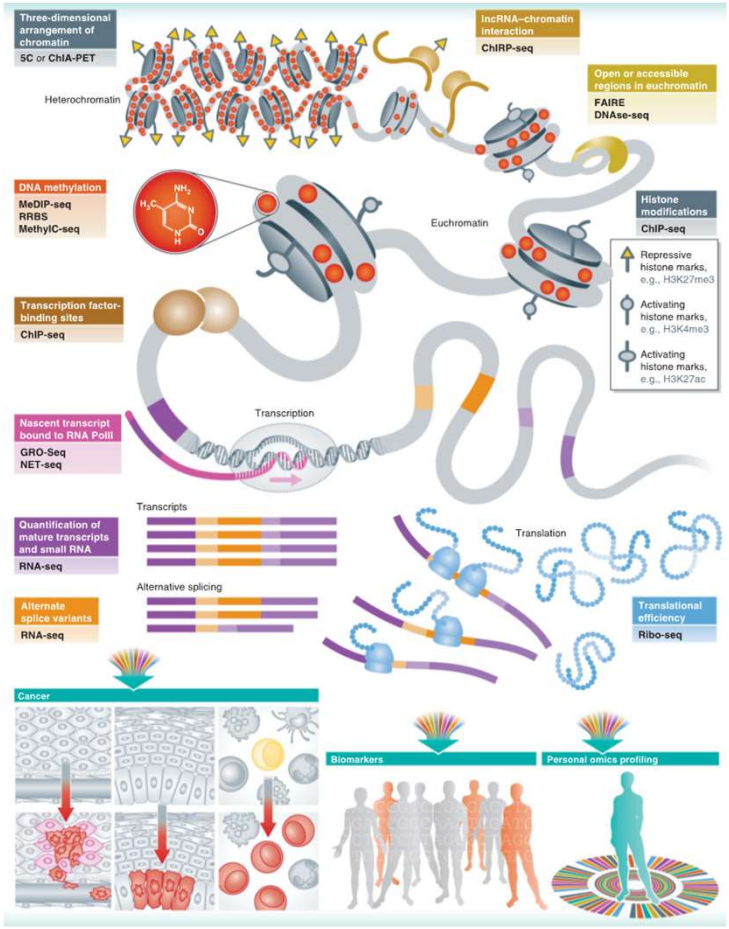
Ion Proton



PacBio

° ° °

Wide Applications of HTS





ENCODE

International Cancer Genome Consortium

GENOME 10K

i5k

HMP

earth microbiome project